



CN_Group11_ Impact of Vehicle Speed on Handover and Packet Loss

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CONTENTS

01 — Key Concepts

- V2X and V2I Concepts
- Handover and Packet Loss Concepts

02 — Project Background

03 — Simulation

- Overview and Tools

04 — Simulation

Environment and Test Video

05 — Results

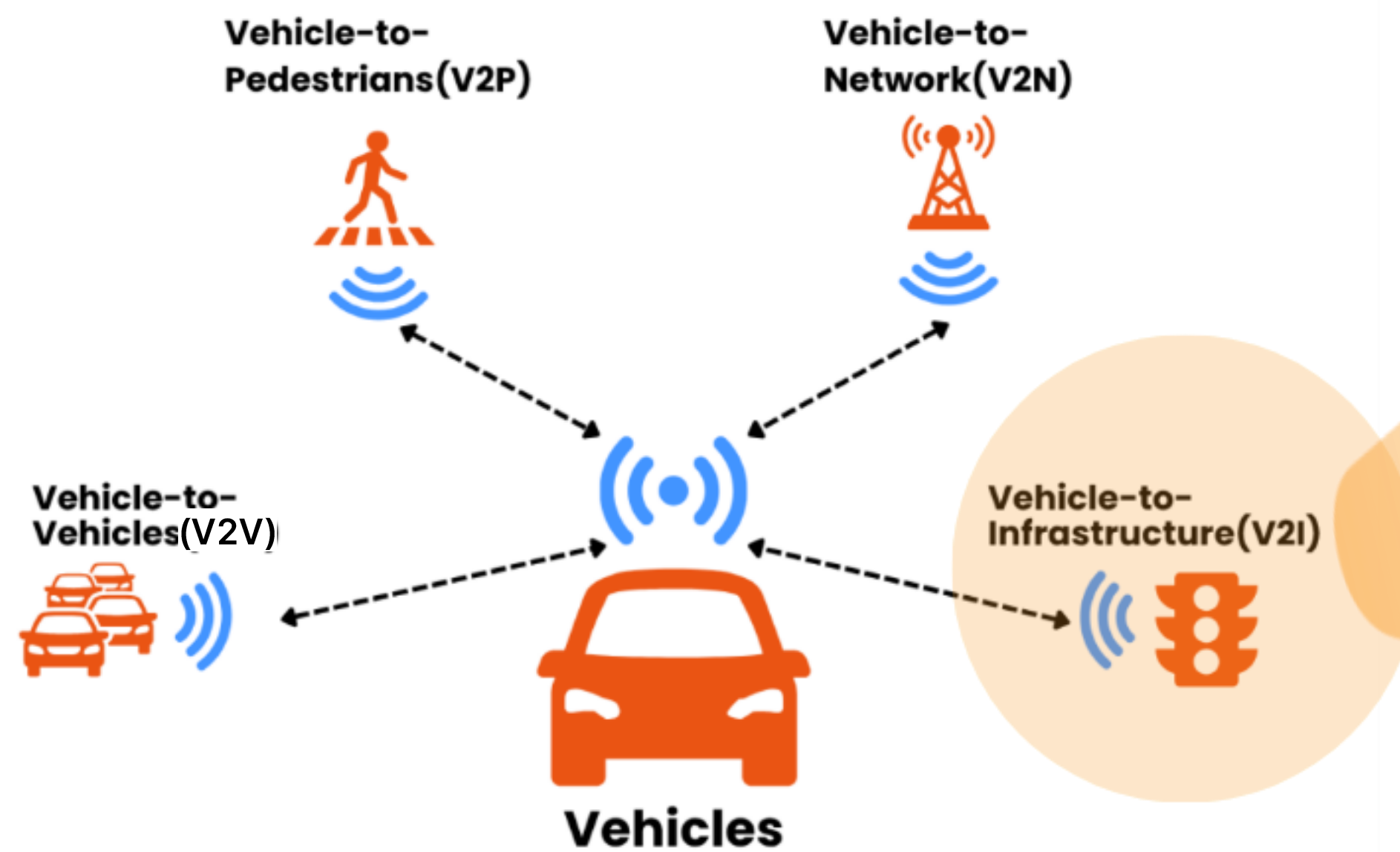
- Handover-like Events vs. Time
- Lost Packets vs. Time
- Limitations

06 — Paper Comparison and Validation

01 Key Concepts V2X/V2I

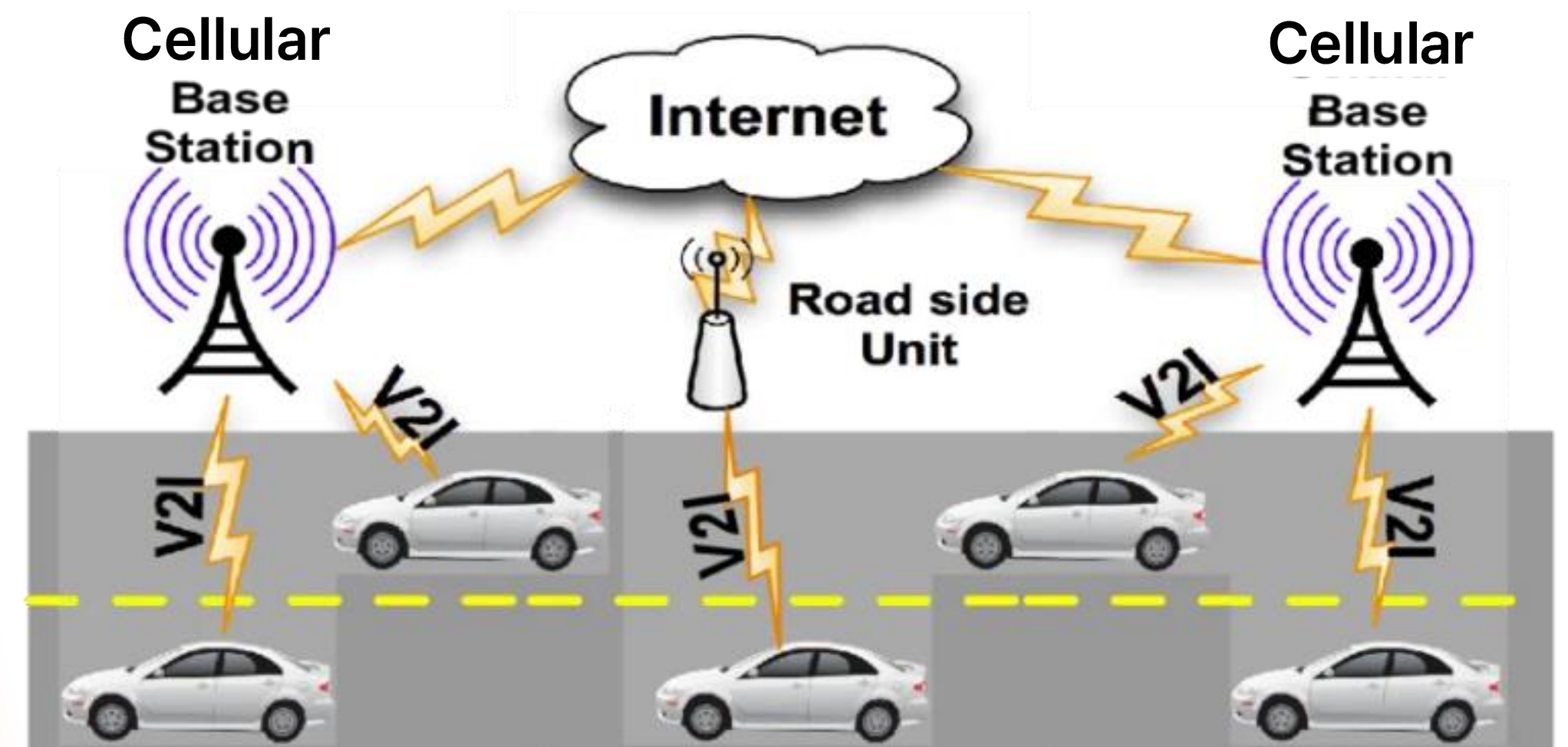
1. V2X?

A vehicle communication environment where vehicles exchange data with various surrounding entities.



2. V2I?

A communication method in which vehicles exchange data with roadside RSUs or base stations.



01 Key Concepts

Handover / Packet Loss

3. handover

A process where a moving vehicle changes its connection from one RSU to another.

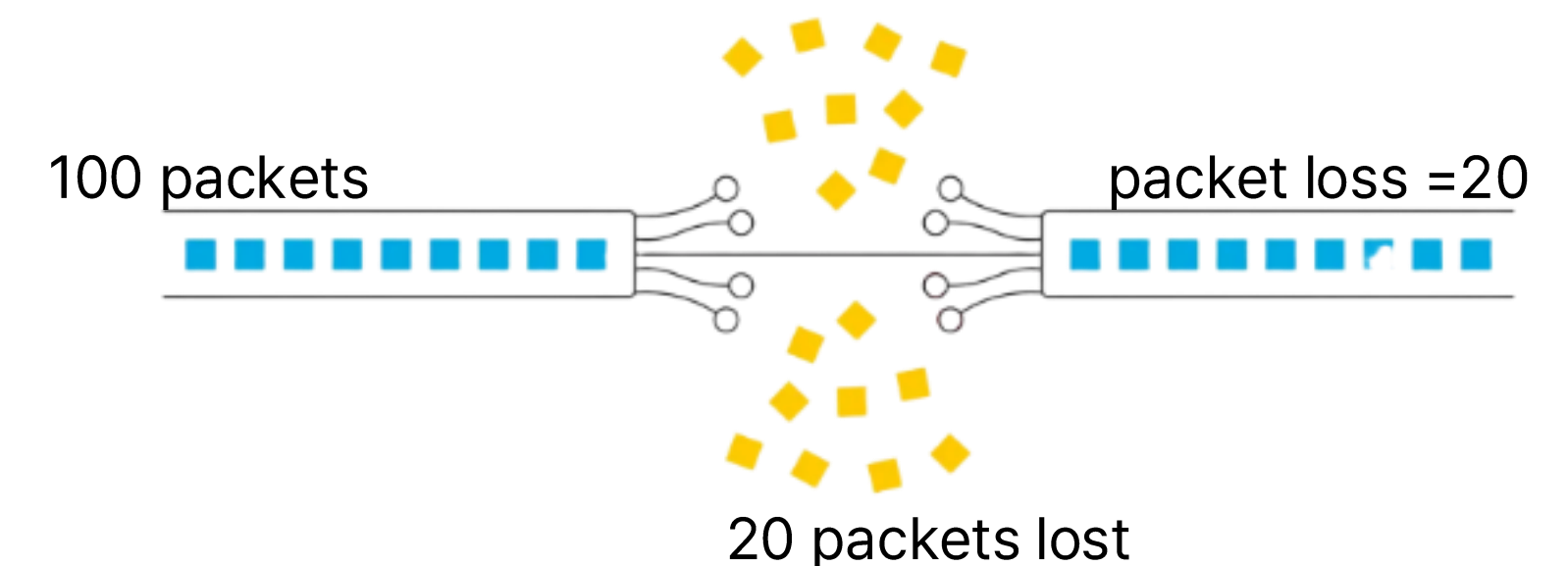
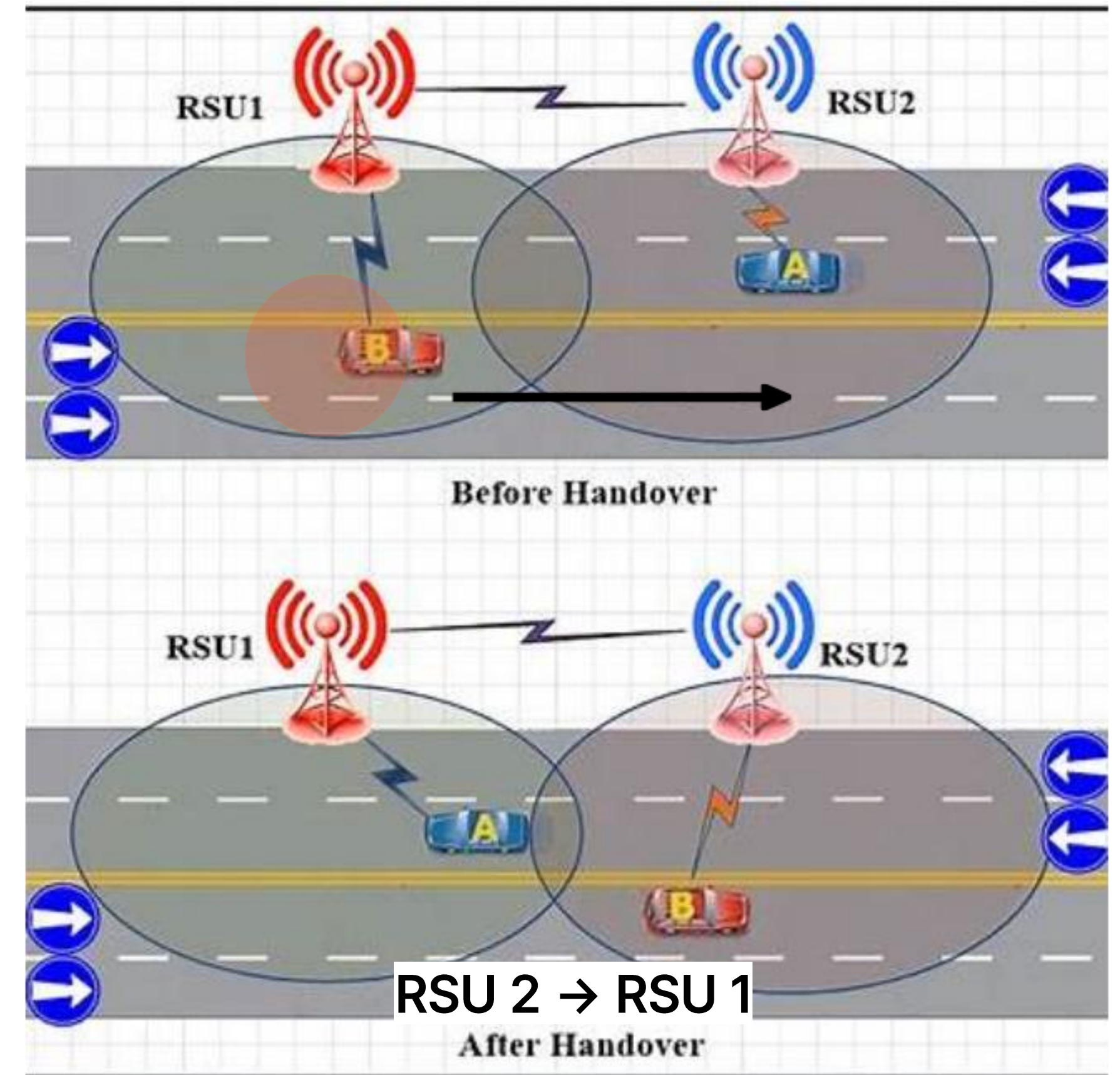
- As vehicle speed increases, the vehicle passes through RSU coverage areas more quickly.
- As a result, handover-like events occur more frequently.

In this project, a handover-like event was recorded whenever the nearest connected RSU changed.

4. packet loss

A situation where some transmitted data packets fail to reach the destination.

- Higher speed and longer distance increase the probability of packet loss.
- Therefore, high-speed vehicles tend to have more lost packets.



Project Background

✍ How stable is V2I communication in a high-speed mobility environment?

→ *Faster vehicles switch RSU connections more often, which can make communication quality less stable.*

1. Importance of V2I Communication

Autonomous vehicles and connected cars continuously exchange data with road infrastructure.

2. RSU-based Connection

Vehicles receive traffic information, hazard alerts, and driving support data through RSUs.

3. Increased Handover

Higher vehicle speed can cause more frequent RSU changes.

4. Communication Quality Issues

packet Frequent handovers may increase packet loss and delay.

5. Need for Analysis

This project analyzes how vehicle speed affects handover and packet loss.



03

Simulation Overview and Tools

- ✍ This project does not implement an actual LTE/5G protocol.
Instead, it is a simplified simulation based on vehicle movement data.

→ *The main focus is to analyze how changes in vehicle speed affect handover and packet loss.*

Tools Used



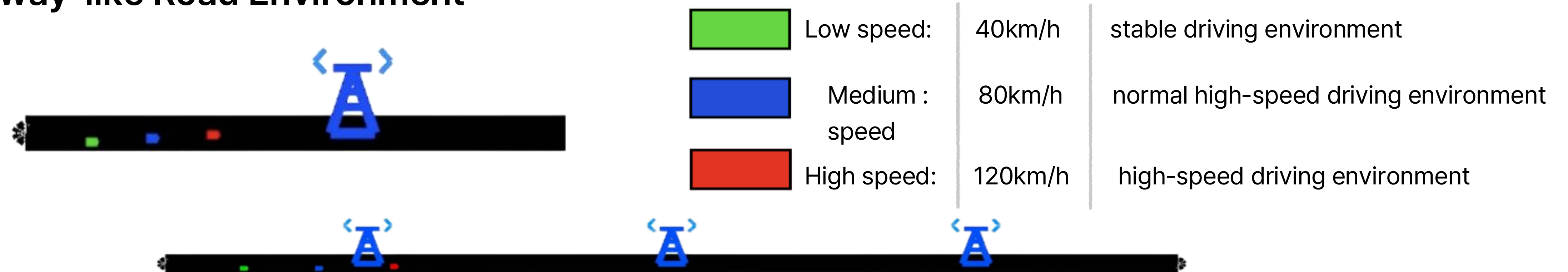
- Constructed a highway-like road environment
- Simulated vehicle traffic movement
- Set different vehicle speed conditions
 - Low speed: 40 km/h
 - Medium speed: 80 km/h
 - High speed: 120 km/h

- Checked vehicle position and speed data
- Calculated the distance between each vehicle and RSU
- Determined the connected RSU
 - Calculated handover-like events
 - Calculated packet loss

04

Simulation Environment

Highway-like Road Environment



Number of RSUs: 3

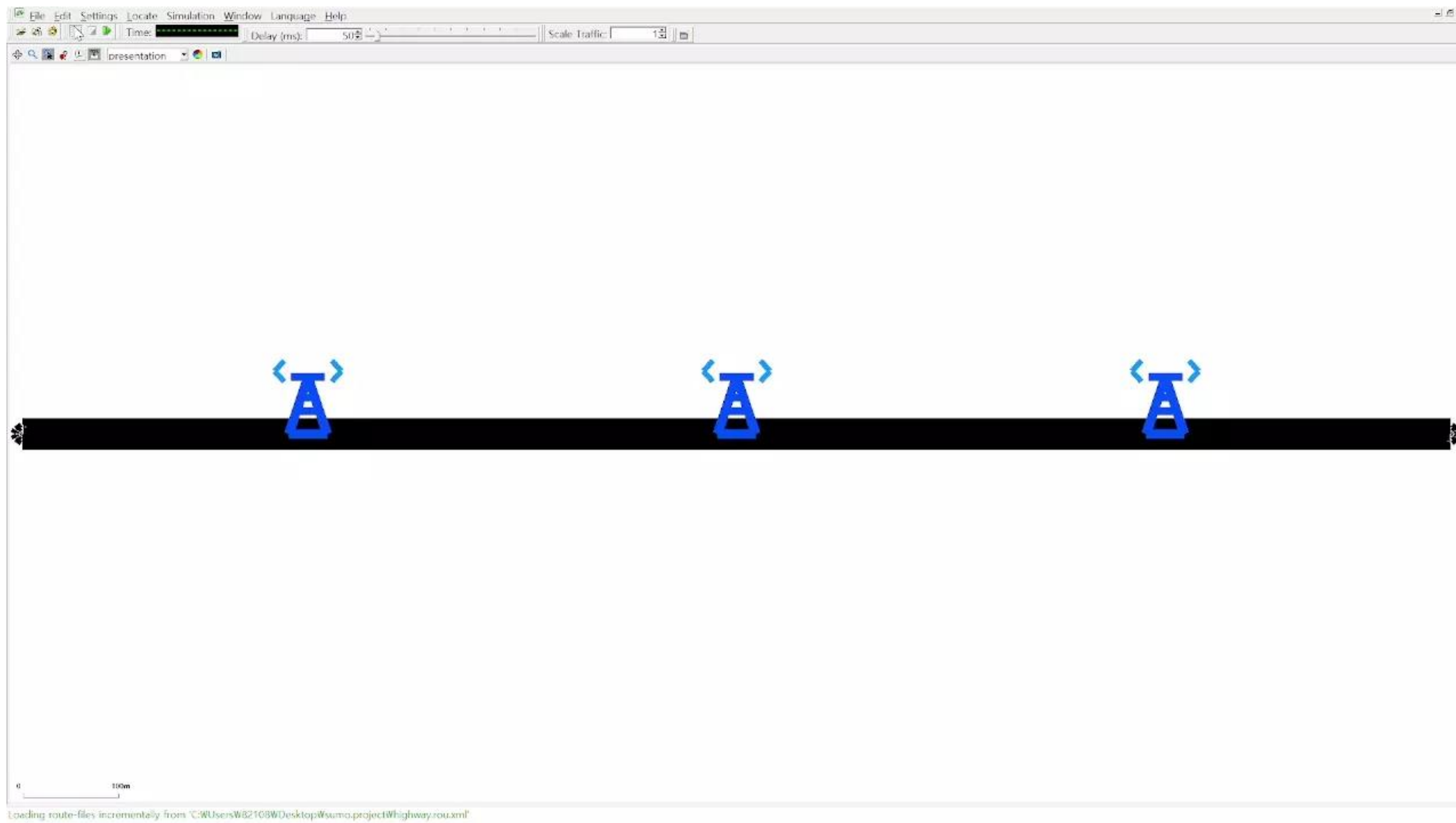
Connection rule: Each vehicle is connected to the nearest RSU

Recording rule: A handover-like event is recorded when the connected RSU changes

Packet Loss Modeling

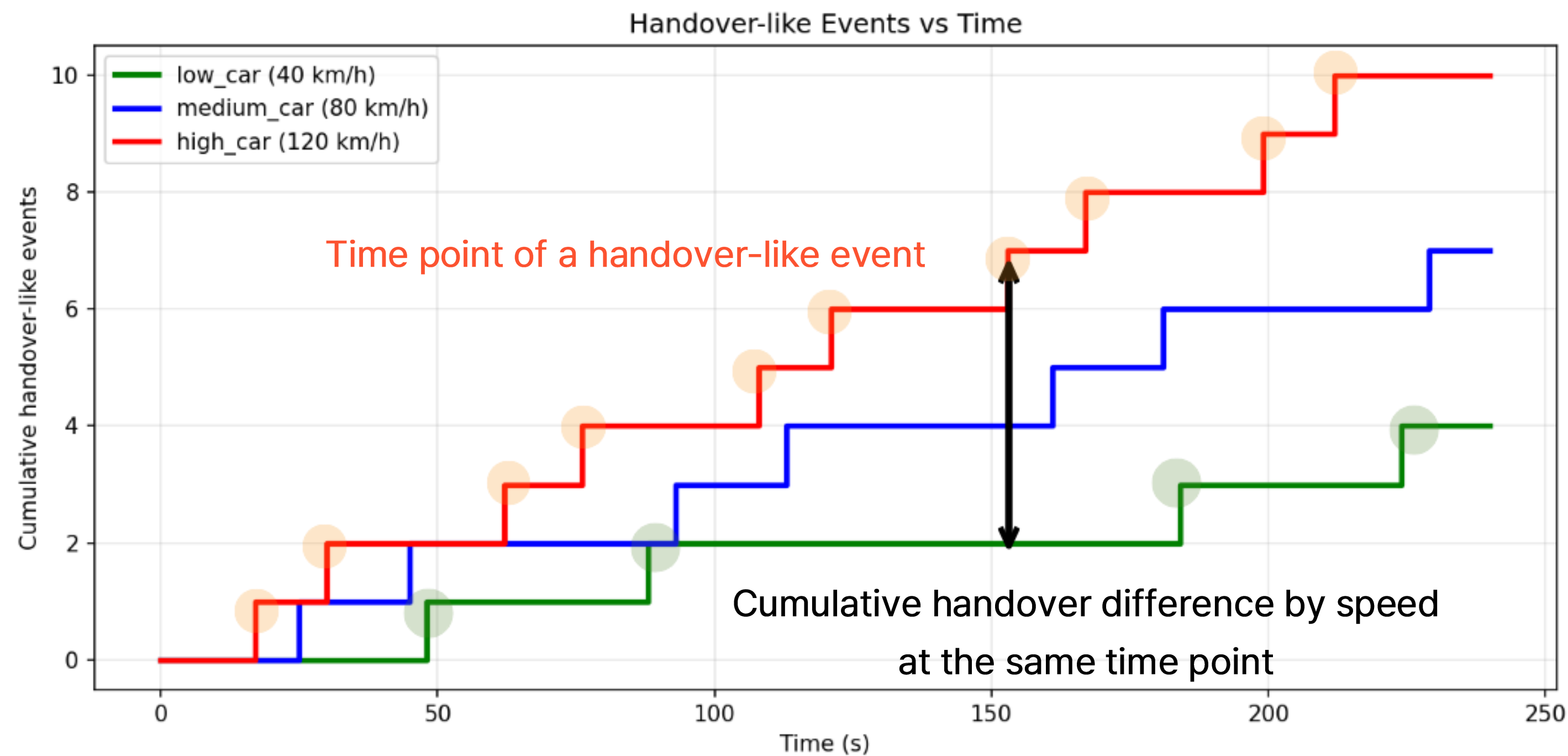
- Vehicles send packets every 1 second
 - Packet loss probability increases as distance from the RSU increases
 - Packet loss probability also increases as vehicle speed increases
- A simplified distance- and speed-based probability model was used

04 Simulation Video



05

Handover-like Events vs Time



X-axis: Simulation time (s)

Y-axis: Cumulative number of handover-like events

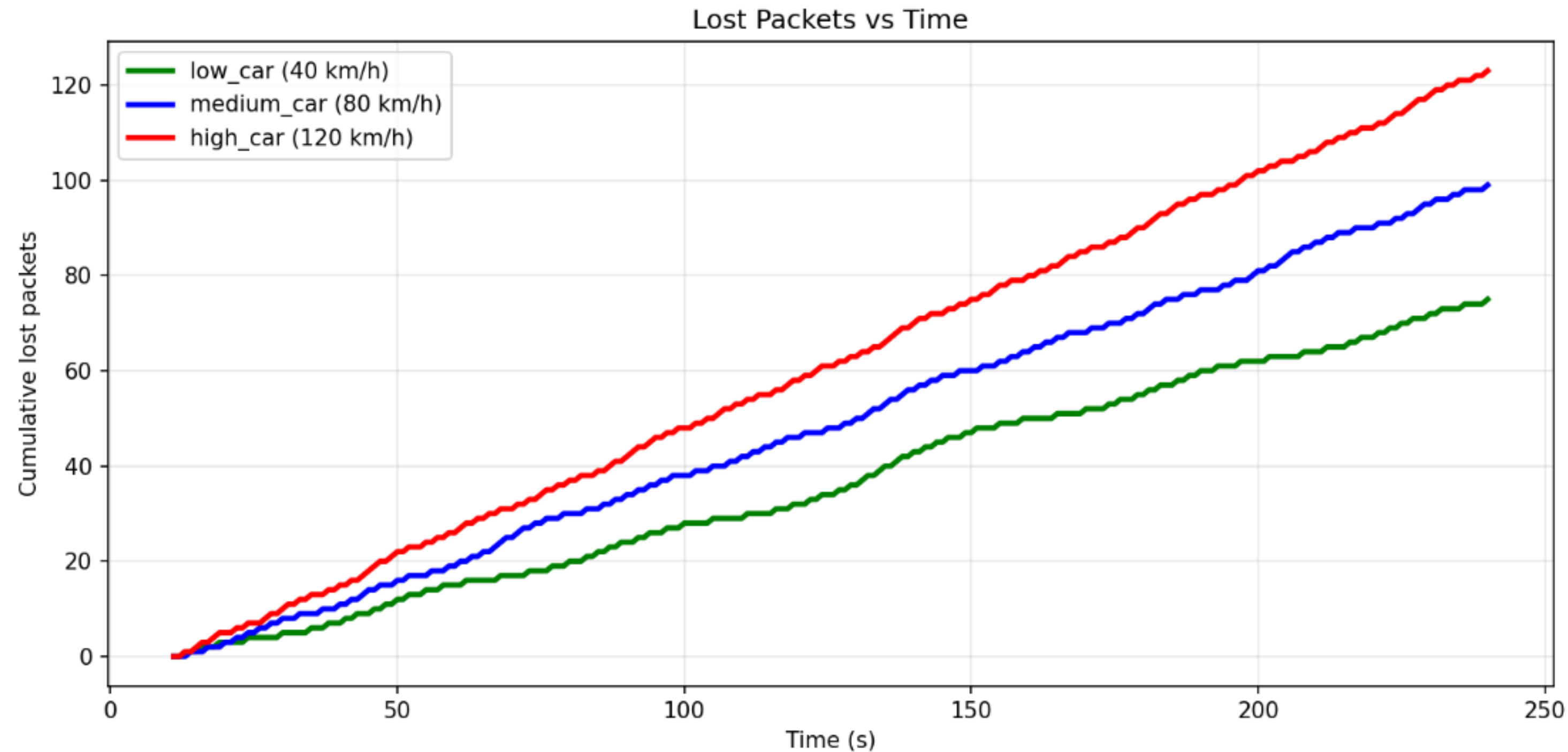
Low speed:	40km/h	stable driving environment
Medium : speed	80km/h	normal high-speed driving environment
High speed:	120km/h	high-speed driving environment

- As vehicle speed increases, the vehicle passes through each RSU coverage area in a shorter time.
- As a result, the connected RSU changes more frequently, leading to more handover-like events.
- At the same simulation time, the high-speed vehicle shows a larger cumulative number of handover-like events than the low-speed vehicle.

Higher Speed → Shorter RSU Dwell Time → More Frequent Connection Switching

05

Lost Packets vs Time



X-axis: Simulation time (s)

Y-axis: Cumulative number of lost packets

Low speed:	40km/h	stable driving environment
Medium : speed	80km/h	normal high-speed driving environment
High speed:	120km/h	high-speed driving environment

- Final cumulative packet loss:
High-speed: ~123 > Medium-speed: ~99
> Low-speed: ~75
- Faster vehicles have a shorter stable connection time with RSUs.
- Rapid distance changes from RSUs make the connection less stable, leading to increased packet loss.

Higher Speed → Lower Connection Stability → More Packet Loss

05 Limitations

This result does not represent actual LTE/5G performance, but shows a simplified trend in a high-speed mobility environment.

1. Actual Handover Protocol Not Implemented

In this project, a handover-like event was recorded when the connected RSU changed. However, actual LTE/5G handover involves more complex factors, such as:

- RSRP: Reference Signal Received Power
- SINR: Signal-to-Interference-plus-Noise Ratio
- Handover trigger conditions
- Time-to-Trigger: The required time for a trigger condition to be maintained
- Base station interfaces: X2, NG, etc.

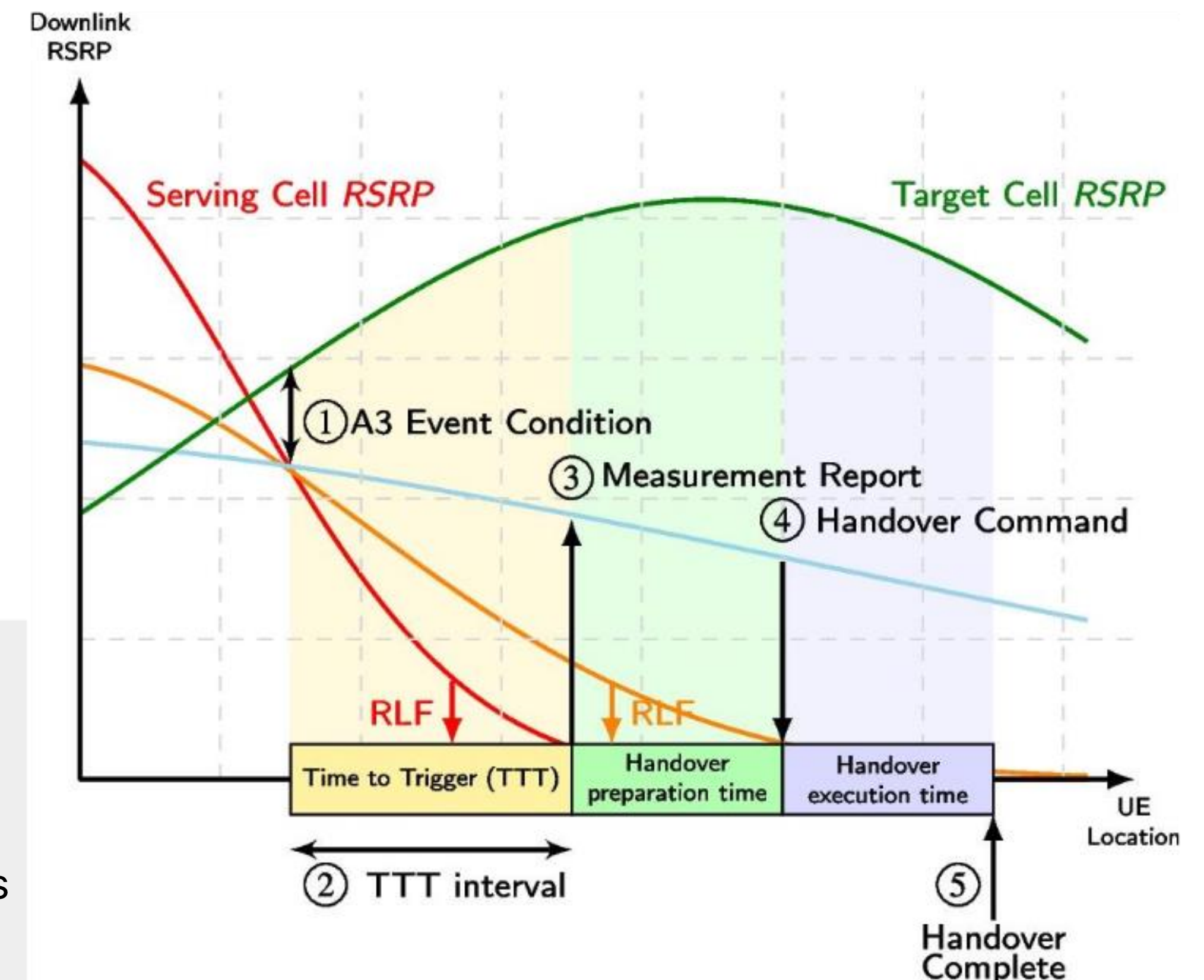
2. Simplified Packet Loss Model

Packet loss in this project was calculated using a simple distance- and speed-based probability model, not an actual wireless channel model.

In real environments, packet loss can also be affected by:

- Radio wave interference
- Obstacles
- Terrain
- Wireless channel conditions
- Weather and surrounding traffic density

Actual handover requires a more complex network procedure, including RSRP, SINR, trigger conditions, and time-to-trigger.



06 Paper Comparison and Validation

The main trends are consistent with previous studies.

Paper	Research Focus	Connection to Our Results
C-V2X Mode 4, 2022	PDR analysis at 40 / 80 / 120 km/h	<u>Supports the reliability of the speed conditions used in our simulation</u>
LTE Handover, 2018	Handover analysis at 3–170 km/h	<u>Supports that higher speed leads to more handovers</u>
VANET AODV, 2022	PDR, delay, and overhead at 60 / 80 km/h	Confirms PDR decrease and packet loss increase
IEEE 802.11p, 2012	Impact of mobility on communication performance	<u>Supports that mobility can degrade communication quality</u>
Sensors, 2022	Real-road test with LTE-5G devices	<u>Shows that distance and speed affect real communication performance</u>

Conclusion:



Higher Vehicle Speed Reduces V2I Communication Stability

This project provides a basic analysis showing why communication stability is important in high-speed mobility environments.

1. What we did

We implemented a simplified high-speed V2I environment using SUMO and Python.



2. What we found

Higher speed → More handover-like events

Higher speed → More packet loss

In high-speed vehicles, RSU connections changed more frequently and packet loss became larger.

3. Why it matters

Since vehicles pass through RSU coverage areas quickly at high speeds, designing stable V2I communication is essential.

4. Limitation

Since actual LTE/5G protocols were not fully implemented, the results are closer to a trend analysis than an exact performance prediction.



Thank you.